Q3. Wireshark Capture and MAC Address Extraction [Mark: 1.0]

A capture was taken on a wired interface of a lab PC with the display filter `eth.type or arp`, to observe physical addressing, protocol multiplexing and broadcast address resolution. **Host:** 192.168.10.25 (3C:52:82:A1:07:9B). **Default gateway:** 192.168.10.1 (9C:A2:F4:5D:30:11).

No.	Time (s)	Source MAC	Destination MAC	Protocol	Info
1	0.000000	3C:52:82:A1:07:9B	FF:FF:FF:FF:FF:FF	ARP	Who has 192.168.10.1? Tell 192.168.10.25
2	0.000980	9C:A2:F4:5D:30:11	3C:52:82:A1:07:9B	ARP	192.168.10.1 is at 9C:A2:F4:5D:30:11
3	0.012400	3C:52:82:A1:07:9B	9C:A2:F4:5D:30:11	DNS/UDP	Standard query A example.org
4	0.031750	3C:52:82:A1:07:9B	9C:A2:F4:5D:30:11	TCP	51234 -> 443 [SYN] Seq=0
5	0.058300	9C:A2:F4:5D:30:11	3C:52:82:A1:07:9B	TCP	443 -> 51234 [SYN, ACK]

Figure 2.1: Ethernet_Capture_Trace.pcapng (Wireshark filter: eth.type or arp), 5 packets.

Frame	EtherType	Layer 2 analysis
1	0x0806 (ARP)	Broadcast so every device on the segment sees it; the gateway recognises its own IP and answers.
2	0x0806 (ARP)	Unicast reply telling the host the gateway's MAC address; the host caches it.
3	0x0800 (IPv4)	The DNS server is off-subnet, so the frame is addressed to the gateway's MAC while the IP destination is the DNS server.
4	0x0800 (IPv4)	First step of the TCP handshake; the destination MAC is again the gateway, not the remote server.
5	0x0800 (IPv4)	The gateway re-frames the server's reply with the host's MAC as destination.

Observation: MAC addresses change at every hop, while IP addresses stay end-to-end. Off-subnet traffic is always addressed at Layer 2 to the gateway.

Q4. Switch MAC Learning and Forwarding [Mark: 1.0]

A switch is a multi-port Layer 2 device that builds a **CAM (MAC address) table** and uses it to forward frames only where needed.

1. **Learning:** for every incoming frame, record *source MAC -> ingress port* with an ageing timer (commonly 300 s).
2. **Flooding:** if the destination is broadcast or unknown, send the frame out of every port except the ingress port.
3. **Forwarding / filtering:** once the destination is known, send the frame only to that port.

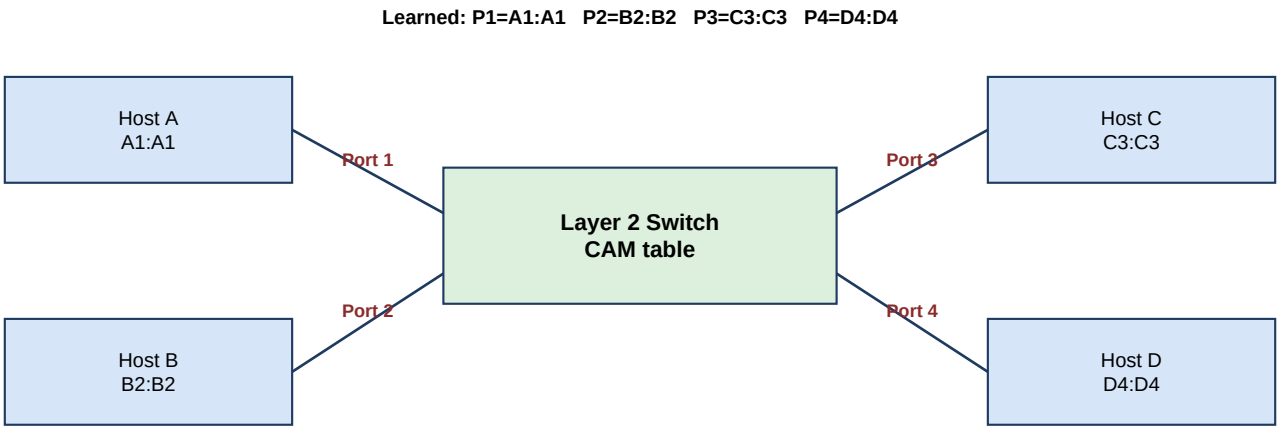

Figure 3.1: Switch with four hosts; CAM table learned from source addresses.

Event	Table after event	Action
A1:A1 -> C3:C3	A1:A1 = Port 1	C3:C3 unknown, so flood to ports 2, 3, 4
C3:C3 -> A1:A1	C3:C3 = Port 3	A1:A1 known, forward only to port 1
B2:B2 -> A1:A1	B2:B2 = Port 2	Forward only to port 1
D4:D4 -> FF:FF:FF:FF:FF:FF	D4:D4 = Port 4	Broadcast, flood to ports 1, 2, 3

Q5. Hub vs Bridge vs Switch [Mark: 1.0]

Attribute	Hub	Bridge	Switch
OSI layer	Layer 1 (Physical)	Layer 2 (Data-Link)	Layer 2 (multilayer models also act at Layer 3)
Collision domains	One shared by all ports	One per port (few ports, typically 2-4)	One per port
Broadcast domains	One	One	One per VLAN (default: one)
Forwarding method	Repeats bits to every other port	Software MAC lookup, store-and-forward	Hardware (ASIC) lookup at wire speed
Duplex	Half only	Mostly half	Full duplex
Use today	Obsolete	Legacy	Standard in campus and enterprise LANs

Q6. CSMA/CD and Why It Is Unused in Modern Ethernet [Mark: 1.0]

How it worked. On shared media (10BASE5, 10BASE2, hubs) a station first **listened** to the channel and transmitted only if it was idle. If two stations transmitted together, they detected the collision, stopped, sent a 32-bit **jam signal** and waited a random time chosen by **truncated binary exponential backoff**: after the k-th collision wait r x slot time, where $0 <= r < 2^k$ (k capped at 10, frame abandoned after 16 attempts).

Why it is no longer used. Modern Ethernet (100BASE-TX, 1000BASE-T, fibre) connects each device to its own switch port by a point-to-point link with separate transmit and receive paths running in full duplex. Because no two stations compete for the same channel, collisions cannot occur and collision detection has nothing to do. The standard still defines CSMA/CD for half-duplex links, but in practice it is disabled.

Q7. IEEE 802.11 CSMA/CA vs CSMA/CD [Mark: 1.0]

A wireless station cannot listen while it transmits, because its own signal swamps its receiver, so collisions cannot be detected. Wi-Fi therefore **avoids** collisions instead.

Feature	CSMA/CD (802.3)	CSMA/CA (802.11)
Approach	Detect a collision and abort	Avoid collisions before they happen
Channel sensing	Electrical signal on the wire	Physical carrier sense (clear channel assessment) plus virtual carrier sense through the NAV timer
Waiting rule	Fixed inter-packet gap (96 bit times)	Inter-frame spaces of different lengths (SIFS shortest, then PIFS, DIFS) plus random backoff
Handshake	None	Optional RTS/CTS to handle the hidden-node problem
Acknowledgement	Not needed at Layer 2	Every unicast frame must be acknowledged; a missing ACK triggers retransmission

Q8. Wireless LAN Design for a Seminar Hall [Mark: 1.0]

Facility: seminar hall of 24 m x 15 m with up to 90 simultaneous users (laptops and phones). **Access points:** three dual-band IEEE 802.11ax (Wi-Fi 6) ceiling APs spaced evenly along the long axis at about 3 m height, giving roughly 30 clients per AP, with transmit power reduced so cells overlap only slightly.

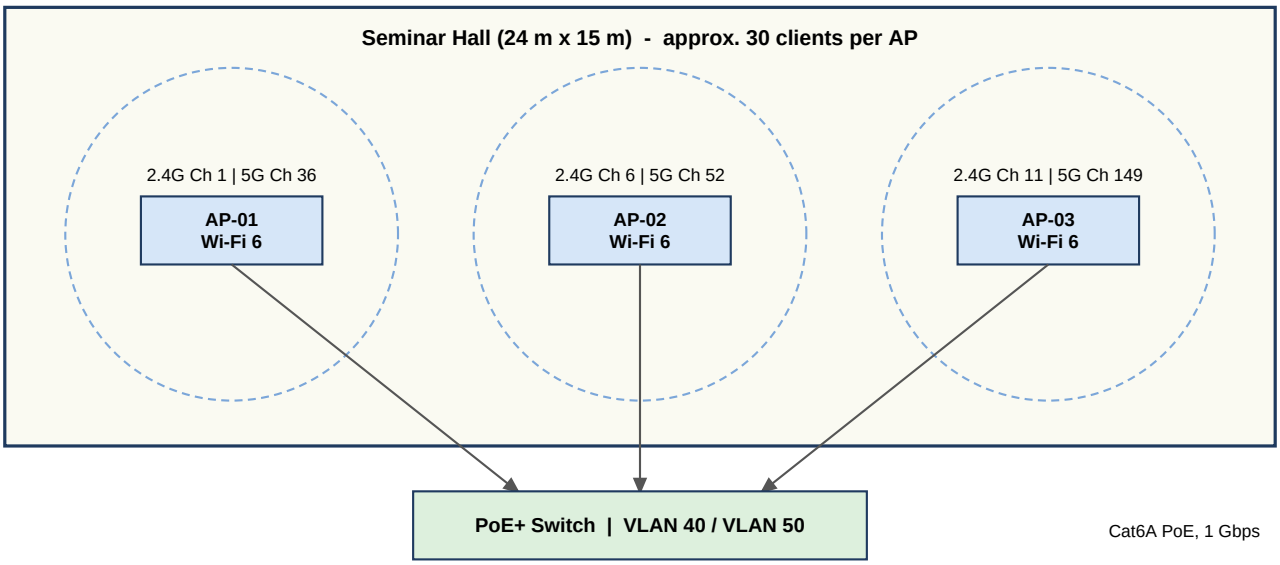

Figure 4.1: Three-AP layout with non-overlapping channels and a PoE+ switch uplink.

Item	Design choice
2.4 GHz channels	1, 6 and 11 (20 MHz) - the three non-overlapping channels, one per AP
5 GHz channels	36, 52 and 149 (40 MHz), one per AP, to avoid co-channel interference
Cabling and power	Cat6A to a PoE+ switch, 1 Gbps uplinks
SSIDs and VLANs	`Hall-Guest` on VLAN 40 (internet only); `Hall-Staff` on VLAN 50 (internal resources)
IP addressing	10.30.8.0/23 (510 usable hosts), gateway 10.30.8.1
DHCP lease	1 hour, as users come and go in short sessions
Security	WPA3-Enterprise with 802.1X (EAP-TLS) against a RADIUS server; client isolation enabled so users cannot reach each other's devices

Reasoning. Three channels per band stop neighbouring APs interfering with each other. Separate VLANs keep guest traffic away from staff systems, and the /23 subnet leaves ample address space for 90 users.

Q9. Mobile IP and Mobility Support [Mark: 1.0]

Problem. An IP address contains the network prefix of its subnet. If a device moves to another network and keeps its address, routers still send its packets to the old network; if it takes a new address, its TCP connections break. Mobile IP (RFC 5944 for IPv4) lets the device keep one permanent address while moving.

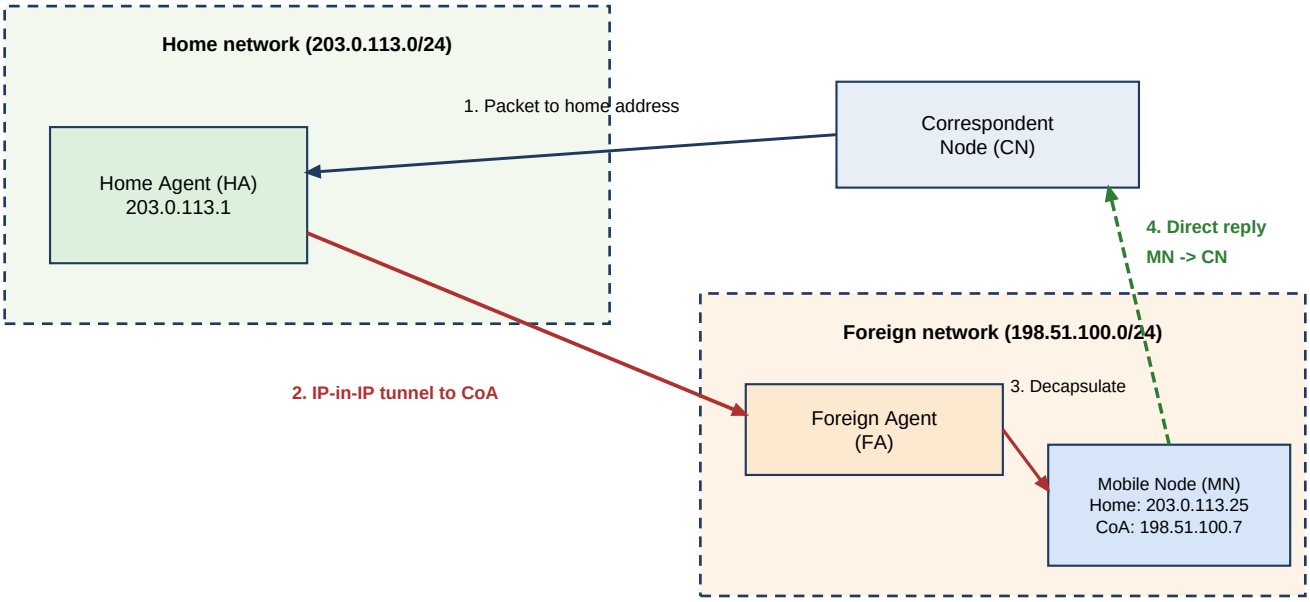

Figure 5.1: Mobile IP - tunnelling via the Home Agent and triangular routing.

- **Mobile Node (MN):** the moving device with a permanent home address (203.0.113.25).
- **Home Agent (HA):** router on the home network that tracks the MN's location (203.0.113.1).
- **Foreign Agent (FA):** router on the visited network that assists the MN.
- **Care-of-Address (CoA):** temporary address marking the MN's present location (198.51.100.7).
- **Correspondent Node (CN):** the peer communicating with the MN.

Operation.

1. **Discovery:** the MN hears agent advertisements and realises it is on a foreign network.
2. **Registration:** the MN registers its CoA with the HA (through the FA), creating a binding *home address -> CoA*.
3. **Forwarding:** the CN sends packets to the home address; the HA intercepts them and tunnels them (IP-in-IP) to the CoA.
4. **Delivery:** the tunnel end removes the outer header and hands the original packet to the MN.
5. **Return path:** the MN replies directly to the CN with its home address as source. The forward path goes via the HA and the return path does not, hence **triangular routing**.

Because the MN's address never changes, higher-layer sessions survive the move; the drawback is the longer forward path through the HA.

Q10. Packet-Level Comparison: Ethernet II vs IEEE 802.11 Frame [Mark: 1.0]

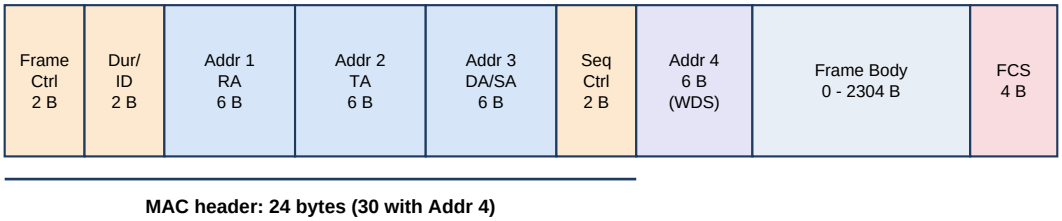

Figure 6.1: IEEE 802.11 data frame layout (RA = receiver, TA = transmitter, DA/SA = destination/source).

Feature	Ethernet II (802.3)	IEEE 802.11
Header size	14 bytes fixed (6 + 6 + 2)	24 bytes typical, 30 with the fourth address (plus 2 more for QoS)
Addresses	2 (destination, source)	Up to 4 (receiver, transmitter, destination, source), because frames pass through an access point or wireless distribution system
Control information	None in the MAC header	Frame Control (2 B): type, subtype, To-DS, From-DS, retry, power management, more data, protected frame
Duration / ID	Absent	2 B; sets the NAV timer for virtual carrier sensing
Sequence control	Absent	2 B (sequence and fragment numbers) to remove duplicates and reassemble fragments
Maximum payload	1500 bytes	2304 bytes
Acknowledgement	None at Layer 2	Mandatory ACK for unicast frames

Conclusion. The data-link layer adapts to its medium. Wired Ethernet uses a short, simple header and hardware switching over full-duplex links, so collisions have disappeared. IEEE 802.11 needs a larger header, multiple addresses, acknowledgements and CSMA/CA to stay dependable on a shared, error-prone radio channel. Mobile IP extends this by letting a device keep its identity while changing networks.